ECO7101
CHAPTER-26
FACTOR MARKETS

Just as there is a demand for and a supply of a product, there is a demand for and a supply of a factor, or resource, such as labor.

The Demand for a Factor

The demand for factors is a derived demand; that is, it is derived from, and directly related to, the demand for the product that the resources go to produce. If the demand for the product rises, so does the demand for the factors that go into the making of the product. If the demand for the product falls, so does the demand for the factors. For example, if the demand for a university education falls, so does the demand for university professors. If the demand for computers rises, so does the demand for skilled computer workers.

When the demand for a seller's product rises, the seller needs to decide how much more of a factor to buy. Marginal revenue product (MRP) and marginal factor cost (MFC) are relevant to this decision.

Marginal revenue product (MRP) is the additional revenue generated by employing an additional factor unit, such as one more unit of labor. For example, if a firm employs one more unit of a factor and its total revenue rises by \$20, the MRP of the factor equals \$20. Marginal revenue product can be calculated in two ways, either as

$$MRP = \frac{\Delta TR}{\Delta \text{Quantity of the factory}}$$

or as

$$MRP = MR \times MPP$$

where TR = total revenue, MR = marginalrevenue, and MPP = marginal physical product.

Exhibit 1 shows the two methods for calculating MRP for a hypothetical firm.

EXHIBIT 1

Calculating Marginal Revenue Product (*MRP*)

There are two methods of calculating *MRP*. Part (a) shows one method ($MRP = \Delta TR/\Delta$Quantity of the factor), and part (b) shows the other ($MRP = MR \times MPP$).

(1) Quantity of Factor X	(2) Quantity of Output, *Q*	(3) Product Price, Marginal Revenue ($P = MR$)	(4) Total Revenue $TR = P \times Q$ = (3) × (2)	(5) Marginal Revenue Product of Factor X: $MRP = \Delta TR/\Delta$Quantity of Factor X = $\Delta(4)/\Delta(1)$
0	10*	$5	$50	–
1	19	5	95	$45
2	27	5	135	40
3	34	5	170	35
4	40	5	200	30
5	45	5	225	25

(a)

Method: MRP = MR x MPP or MP

(1) Quantity of Factor X	(2) Quantity of Output, *Q*	(3) Marginal Physical Product $MPP = \Delta(2)/\Delta(1)$	(4) Product Price, Marginal Revenue ($P = MR$)	Marginal Revenue Product of Factor X: $MRP = MR \times MPP$ = (4) × (3)
0	10*	–	$5	–
1	19	9	5	$45
2	27	8	5	40
3	34	7	5	35
4	40	6	5	30
5	45	5	5	25

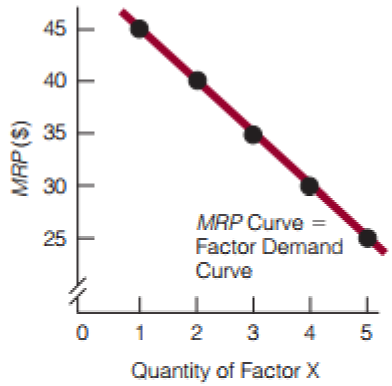
(b)

Both methods derive same marginal revenue products.

The MRP Curve Is the Firm's Factor Demand Curve

We can derive MRP curve from above tables by using MRP data and quantity of factor data (i.e., by using column-1 and column-5) i.e.,

Qty of factor	MRP
0	-
1	45
2	40
3	35
4	30
5	25



The data in columns (1) and (5) in Exhibit 1 are plotted to derive the MRP curve. The curve shows the various quantities of the factor the firm is willing to buy at different prices, which is what a demand curve shows. The MRP curve is the firm's factor demand curve.

Why MRP curve slopes downward?

The MRP curve in Exhibit 2 is downward sloping. You can understand why when you recall that MRP can be calculated as MRP = MR X MPP. With regard to MPP, the marginal physical product of a factor, you know that, according to the law of diminishing marginal returns, eventually the MPP of a factor will diminish. Because MRP is equal to MR X MPP and because MPP will eventually decline, MRP will eventually decline too.

Value Marginal Product (VMP)

Value marginal product (VMP) is equal to the price of the product multiplied by the marginal physical product of the factor:

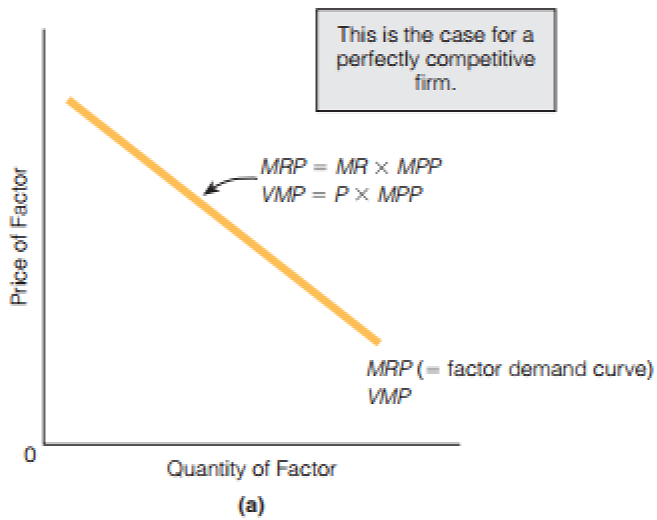
$$VMP = P \times MPP$$

Are MRP and VMP same in perfect and imperfect markets?

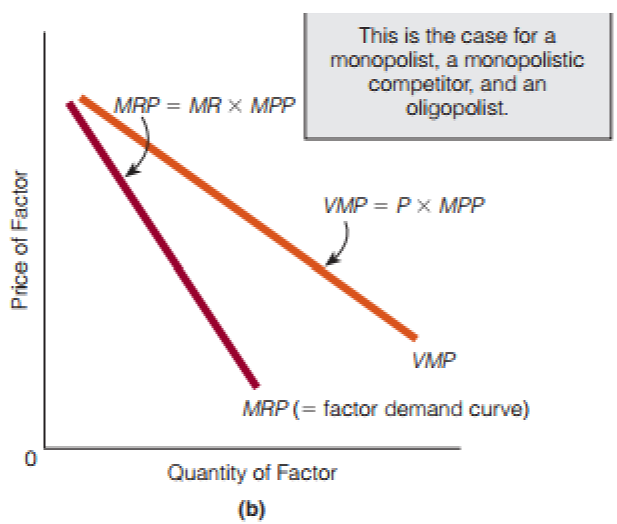
In perfect market: MRP = MR X MPP and VMP = P X MPP.

But in competitive market, P = MR.

That's why MRP = VMP in a competitive market.



In imperfect market, MRP = MR X MPP and VMP = P X MPP.
But, P > MR. Therefore, VMP > MRP and VMP curve lies above MRP curve.



Marginal Factor Cost: The Firm's Factor Supply Curve

Marginal factor cost (MFC) is the additional cost incurred by employing an additional factor unit.

It is calculated as

$$MFC = \frac{\Delta TC}{\Delta \text{Quantity of the factor}}$$

We assumed that the market is a competitive factor market. A firm is factor price taker.

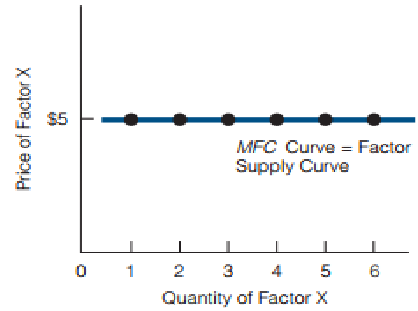
EXHIBIT 4

Calculating *MFC* and Deriving the *MFC* Curve (the Firm's Factor Supply Curve)

In (a), *MFC* is calculated in column 4. Notice that the firm is a factor price taker because it can buy any quantity of factor X at a given price per factor unit ($5, as shown in column 2). In (b), the data from columns (1) and (4) are plotted to derive the *MFC* curve, which is the firm's factor supply curve.

(1) Quantity of Factor X	(2) Price of Factor X	(3) Total Cost $TC = (2) \times (1)$	(4) $MFC = \Delta TC/\Delta$ Quantity of the Factor $= \Delta(3)/\Delta(1)$
0	$5	$0	—
1	5	5	$5
2	5	10	5
3	5	15	5
4	5	20	5
5	5	25	5
6	5	30	5

(a)



(b)

Therefore MRP curve or factor demand curve is downward sloping and MFC or factor supply curve is horizontal.

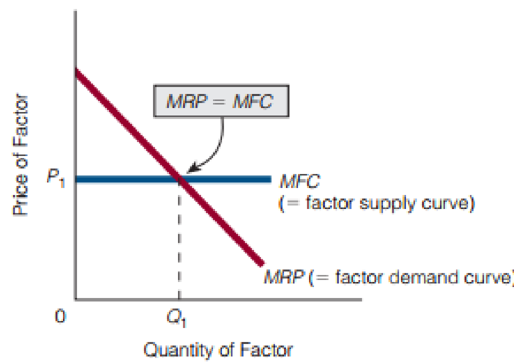
Factor market equilibrium

Equilibrium Condition: MRP = MFC

EXHIBIT 5

Equating *MRP* and *MFC*

The firm continues to purchase a factor as long as the factor's *MRP* exceeds its *MFC*. In the exhibit, the firm purchases Q_1.

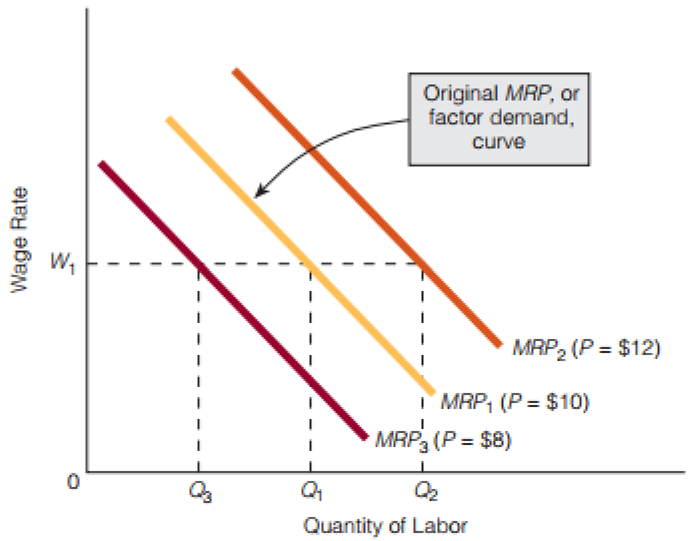


Factor market equilibrium in the diagram is the intersection point MRP and MFC where MRP = MFC. Equilibrium output is Q_1 and equilibrium price P_1. A firm continues buying additional units of the factor until the additional revenue generated by employing an additional factor unit is equal to the additional cost incurred by employing an additional factor unit. Simply stated, keep buying additional units of the factor until MRP = MFC . In Exhibit 5, MRP equals MFC at a factor quantity of Q_1

THE LABOR MARKET

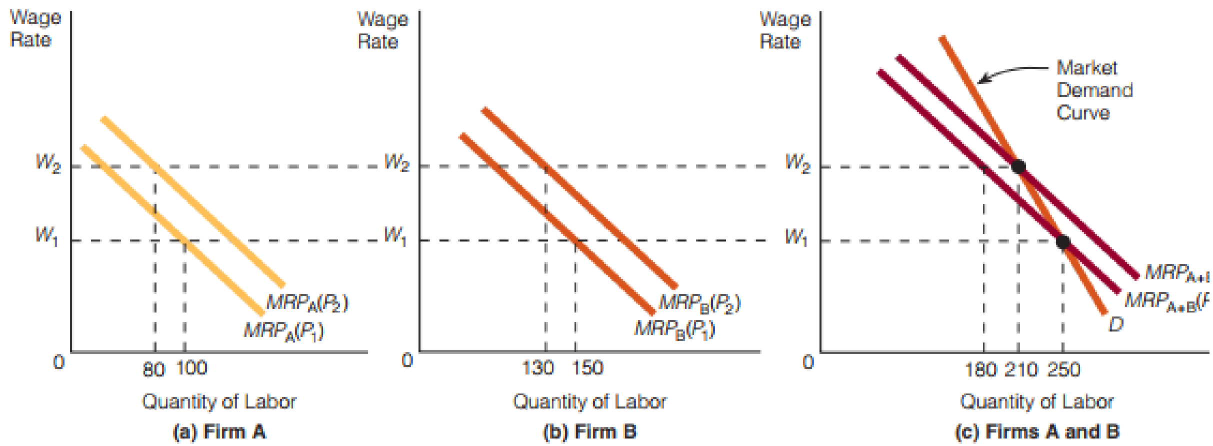
Labor is a factor of special interest because, at one time or another, most people find themselves in the labor market.

What will happen to the factor demand (MRP) curve for labor as the price of the product that the labor produces changes?



In the above diagram, the initial product price is \$10 and the initial factor demand curve is MRP1. At wage rate W_1, the firm hires Q_1 labor. Suppose the product price rises to \$12. As we can see from equation (2), MRP also rises. At each wage rate, the firm wants to hire more labor. For example, at W_1, it wants to hire Q_2 labor instead of Q_1. In short, a rise in product price shifts the firm's MRP, or factor demand, curve rightward. If product price falls from \$10 to \$8, then MRP falls. At each wage rate, the firm wants to hire less labor. For example, at W_1 it wants to hire Q3 labor instead of Q_1. In short, a fall in product price shifts the firm's MRP, or factor demand, curve leftward.

The Derivation of the Market Demand Curve for Labor Units



Two firms, A and B, make up the buying side of the market for labor. At a wage rate of W_1, firm A purchases 100 units of labor and firm B purchases 150 units. Together, they purchase 250 units, as illustrated in (c). Then the wage rate rises to W_2, and the amount of labor purchased by both firms initially falls to 180 units, as shown in (c). Higher wage rates translate into higher costs, a fall in product supply, and a rise in product price from P_1 to P_2. Finally, an increased price raises MRP, and each firm has a new MRP curve. The horizontal addition of the new MRP curves shows that the two firms together purchase 210 units of labor. Connecting the units of labor purchased by both firms at W_1 and W_2 gives the market demand curve.

The Elasticity of Demand for Labor

If the wage rate rises, firms will cut back on the labor they hire. How much they cut back depends on the elasticity of demand for labor, which is the percentage change in the quantity demanded of labor divided by the percentage change in the price of labor (the wage rate). That is,

$$E_L = \frac{\text{Percentage change in quantity demanded of labor}}{\text{Percentage change in wage rate}}$$

For example, when the wage rate changes by 20 percent, the quantity demanded of a particular type of labor changes by 40 percent. The elasticity of demand for this type of labor is, then, 2 (40 percent 4 20 percent), and the demand between the old wage rate and the new wage rate is elastic.

There are three main determinants of elasticity of demand for labor:

- The elasticity of demand for the product that labor produces
- The ratio of labor costs to total costs
- The number of substitute factors

Elasticity of Demand for the Product That Labor Produces

The relationship between the elasticity of demand for the product and the elasticity of demand for labor is as follows:

• The higher the elasticity of demand for the product, the higher is the elasticity of demand for the labor that produces the product.

• The lower the elasticity of demand for the product, the lower is the elasticity of demand for the labor that produces the product.

Ratio of Labor Costs to Total Costs

The relationship between the ratio of labor cost to total cost and the elasticity of demand for labor is as follows:

• The higher the ratio of labor cost to total cost, the higher is the elasticity of demand for labor (i.e., the greater is the cutback in labor for any given wage increase).

• The lower the ratio of labor cost to total cost, the lower is the elasticity of demand for labor (i.e., the less is the cutback in labor for any given wage increase).

Number of Substitute Factors

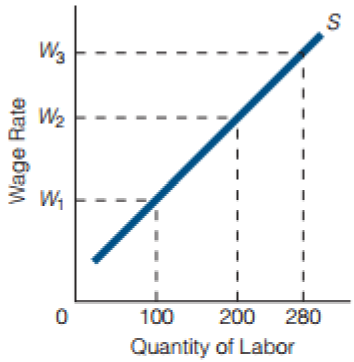
The more substitutes labor has, the more sensitive buyers of labor will be to a change in its price. The more factors that can be substituted for labor, the more likely it is that firms will cut back on their use of labor if its price rises. In sum,

• The more substitutes there are for labor, the higher will be the elasticity of demand for labor.

• The fewer substitutes there are for labor, the lower will be the elasticity of demand for labor.

The market supply curve of labor

As the wage rate rises, the quantity supplied of labor rises, ceteris paribus. The upward-sloping labor supply curve in the following figure illustrates this relationship. At a wage rate of W_1, individuals are willing to supply 100 labor units. At the higher wage rate of W_2,



individuals are willing to supply 200 labor units. Some individuals who were not willing to work at a wage rate of W_1 are willing to work at a wage rate of W_2, and some individuals who were working at W_1 will be willing to supply more labor units at W_2. At the even higher wage rate of W_3, individuals are willing to supply 280 labor units.

Shifts in the Labor Supply Curve

Changes in the wage rate change the quantity supplied of labor units; that is, they cause a movement along a given supply curve. Two factors of major importance, however, can shift the entire labor supply curve: wage rates in other labor markets and the non-money, or non-pecuniary, aspects of a job.

Wage Rates in Other Labor Markets

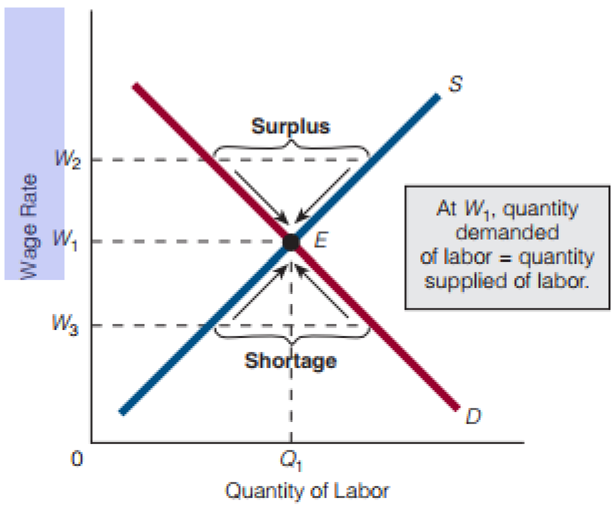
Suppose Deborah works as a technician in a television-manufacturing plant but she has skills suitable for a number of jobs. One day, she learns that the computer-manufacturing plant on the other side of town is offering 33 percent more pay per hour. Deborah is also trained to work as a computer operator, so she decides to leave her current job and apply for work at the computer-manufacturing plant. In short, the wage rate offered in other labor markets can bring about a shift of the supply curve in a particular labor market.

Non-money, or Non-pecuniary

Aspects of a Job Other things held constant, people prefer to avoid dirty, heavy, dangerous work in cold climates. An increase in the overall unpleasantness of a job (e.g., an increased probability of contracting lung cancer by working in a coal mine) will cause a decrease in the supply of labor to the associated firm or industry and a leftward shift in its labor supply curve. An increase in the overall pleasantness of a job (e.g., employees are now entitled to a longer lunch break and use of the company gym) will cause an increase in the supply of labor to the associated firm or industry and a rightward shift in its labor supply curve.

Labor Market Equilibrium

The following diagram illustrates a labor market. The equilibrium wage rate and quantity of labor are established by the forces of supply and demand. At a wage rate of W2, there is a surplus of labor. Some people who want to work at this wage rate will not be able to find jobs and will begin to offer their services for a lower wage rate. The wage rate will move down until it reaches W_1.



At a wage rate of W_3, there is a shortage of labor. Some demanders of labor will begin to bid up the wage rate until it reaches W1. At the equilibrium wage rate, W_1, the quantity supplied of labor equals the quantity demanded of labor.